

Visual preprocessing, sparse codes and navigation accuracy

Expanded ApiaViz simulation study

29 September 2026

Abstract. We tested whether fixed insect-inspired visual processing improves autonomous navigation through a spiking representation. An expanded comparison of 440 trials did not reproduce the accuracy advantage seen in a four-route pilot. ApiaViz and Sobel plus colour each completed 29 of 40 trials; their mean deviations were 55.2 and 50.4 cm. No baseline comparison was significant after multiple-comparison correction. Component tests show that local adaptation makes different places more distinguishable, but this does not establish superior navigation. Both frontends often preserve a heading parallel to the route without steering back towards it.

1 Matched learning and navigation

Insect mushroom bodies learn associations using sparse activity in Kenyon cells (KCs). Ardin et al. [1] applied this principle to route memory; Gattaux et al. [2] combined Sobel filtering with directional memories and robot control. We compare their associated input operations within one shared navigation model, rather than reproducing either complete published system.

Eight approximately 8 m routes in one synthetic grassland were tested with five fixed wiring seeds. Six baselines and five component interventions gave $8 \times 5 \times 11 = 440$ trials. All conditions learned the same nine lateral views spanning ± 20 cm every 10 cm along a route, facing its local tangent. Only image processing changed. Five maps entered the same pooling, standardisation, fixed random projections, 8,000 leaky integrate-and-fire KCs, binary spike-overlap memory and controller. Visual filters were untrained; route acquisition stored the memories. Spike timing had 1 ms resolution.

The agent scanned 13 headings across $\pm 60^\circ$, moved 10 cm towards the most familiar, and stopped within 20 cm of the nest or after 200 steps. There were no corrective resets. This is the project's "open-loop" condition: visual feedback remains available. The eight routes exclude the four-route matched pilot but are not an entirely unexamined project test set.

2 The pilot advantage did not replicate

Input	Completed	Deviation (cm)	Difference (cm)	Holm p
ApiaViz	29/40	55.2	—	—
Grayscale	25/40	63.4	+8.2	1.000
Raw colour	26/40	38.0	-17.2	1.000
Simple opponency	27/40	69.9	+14.7	1.000
Sobel + colour	29/40	50.4	-4.8	1.000
Ardin-style input	6/40	104.7	+49.5	0.195

Mean deviation includes failures. Positive differences favour ApiaViz. Ardin-style input uses inversion, local equalisation and 10×36 downsampling within the shared model.

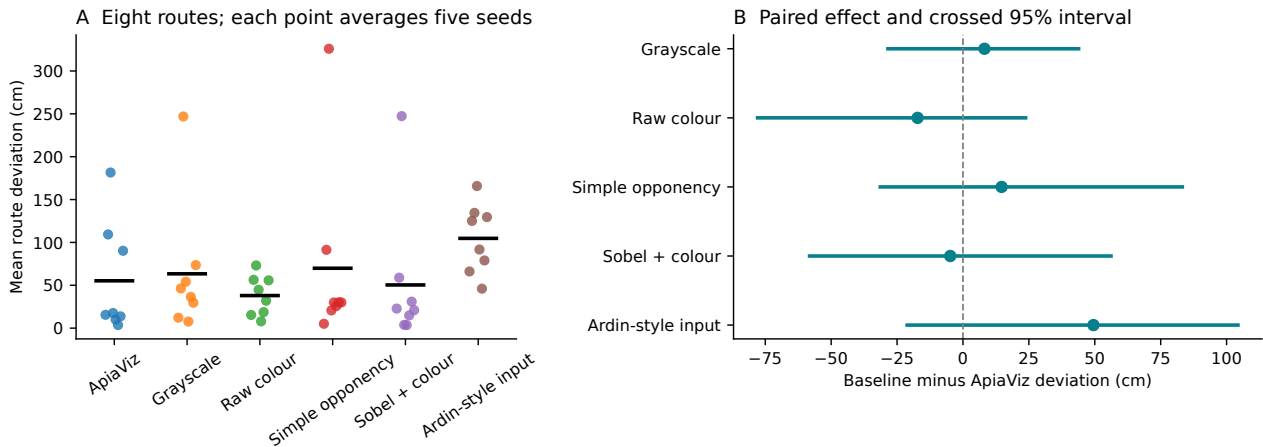


Figure 1: Navigation varies substantially across routes and seeds. Each point averages five seeds on one route. Intervals resample both routes and seeds; all include zero. They are unadjusted 95% intervals.

2 Uncertainty and sensitivity

The primary endpoint was mean distance to the nearest taught route sample over the entire run. The protocol was fixed before inspecting these outcomes. Five seeds were averaged within each route; two-sided sign-flip tests enumerated all 256 sign assignments of the eight route differences. Holm correction covered the five baseline comparisons. These tests assume exchangeable signs under the null and condition on the chosen seeds. Steps and probe images were not treated as independent replicates. Crossed bootstrap intervals used 20,000 resamples of both routes and seeds [5]. All inference remains conditional on one world.

For Sobel minus ApiaViz, the mean difference was -4.8 cm with a crossed 95% interval of $[-58.2, +56.2]$ cm. Seed-specific differences ranged from -65.9 to $+51.0$ cm. The interval is too wide to establish either superiority or equivalence. Ardin-style input had an unadjusted $p = 0.039$, which became $p = 0.195$ after correction.

Long excursions raise the average error, but the early trajectory does not restore the pilot result: mean continuous-route error over the first 50 steps was 16.1 cm for ApiaViz, 15.5 cm for Sobel and 13.9 cm for raw colour. Prespecified secondary analyses using the continuous route or median deviation also gave no significant corrected baseline comparison.

3 What changes inside the representation?

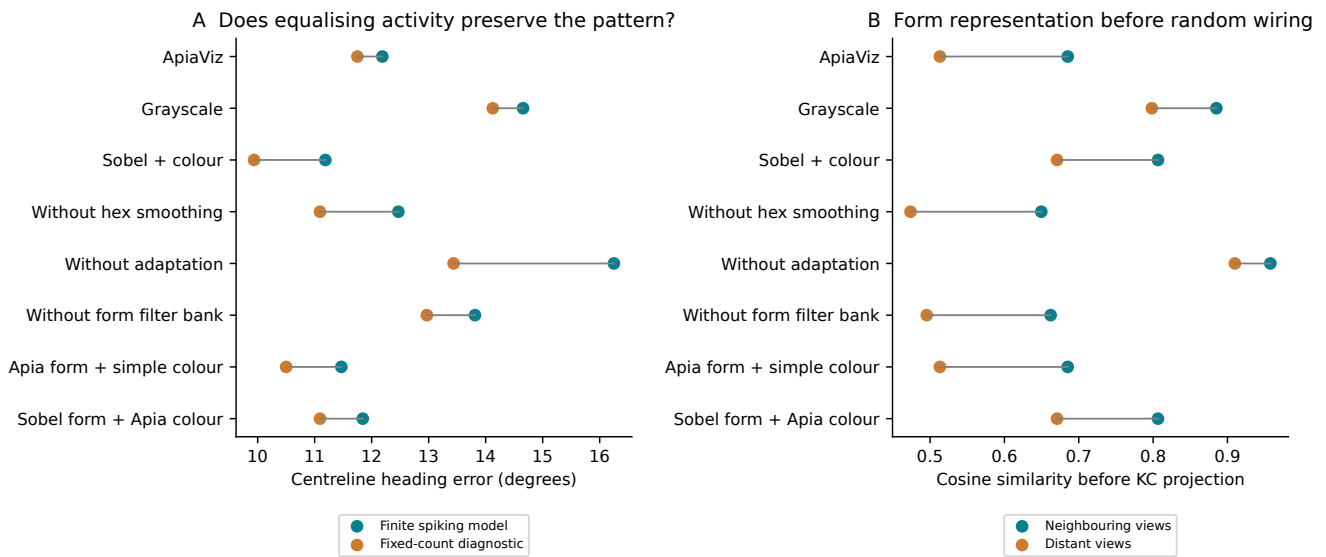


Figure 2: Different representations remain different at equal activity. The offline diagnostic selects exactly 5% of cells per stream from continuous drives, also removing finite timing. The right panel measures form-feature similarity before random wiring. These are descriptive probe results.

Local adaptation reduces ambiguity between places. Removing adaptation raised distant-view form similarity from 0.513 to 0.910 before projection. Effective dimensionality fell from 25.8 to 9.9, indicating less varied features. After spike generation, distant-view overlap rose from 0.432 to 0.611 and centreline heading error increased from 12.2° to 16.3° . This agrees with examining how visual similarity translates into KC recruitment [3]. Nevertheless, the navigation effect of removing adaptation was uncertain: $+19.6$ cm, adjusted $p = 0.719$.

Smoothing trades discrimination for continuity. Removing hexagonal averaging reduced overlap for neighbouring views (0.552 to 0.495) and distant views (0.432 to 0.384). Processed colour also increased similarity at both distances. Neither smoothing removal nor either colour-pathway exchange gave a significant corrected navigation effect.

The form pathway is dominated by smoothed local contrast. About 90% of its standardised squared feature magnitude lay in the smoothed, adapted channel, rather than the two ON/OFF channels. This is not a causal importance estimate. The filter-bank intervention removed all three filters, so it cannot isolate a benefit of centre-surround inhibition.

4 Sparse activity does not guarantee useful steering

We tested identical images at eight positions halfway between teaching samples, five lateral offsets and 13 headings, both clean and at 60% brightness. These common-pose tests produced 35,200 decisions across the main conditions. They separate representation differences from differences in travelled paths.

ApiaViz and Sobel activated 4.92% and 5.08% of KCs on average. Equalising the activity count in 440 additional offline conditions did not reverse their heading-error ranking: ApiaViz gave 11.8°, Sobel 9.9°, and ApiaViz without adaptation 13.4°. Thus local discrimination differences persist at equal counts, but there is still no ApiaViz advantage over Sobel. This diagnostic changes timing as well as count and does not test full navigation.

Uniform darkening changed the chosen heading by 3.0° for ApiaViz, 23.3° without adaptation, and approximately 0.01° for Sobel or raw colour. Shared within-view standardisation already removes uniform gain from homogeneous features. These results support adaptation within this nonlinear pipeline, not superior illumination invariance over simple controls.

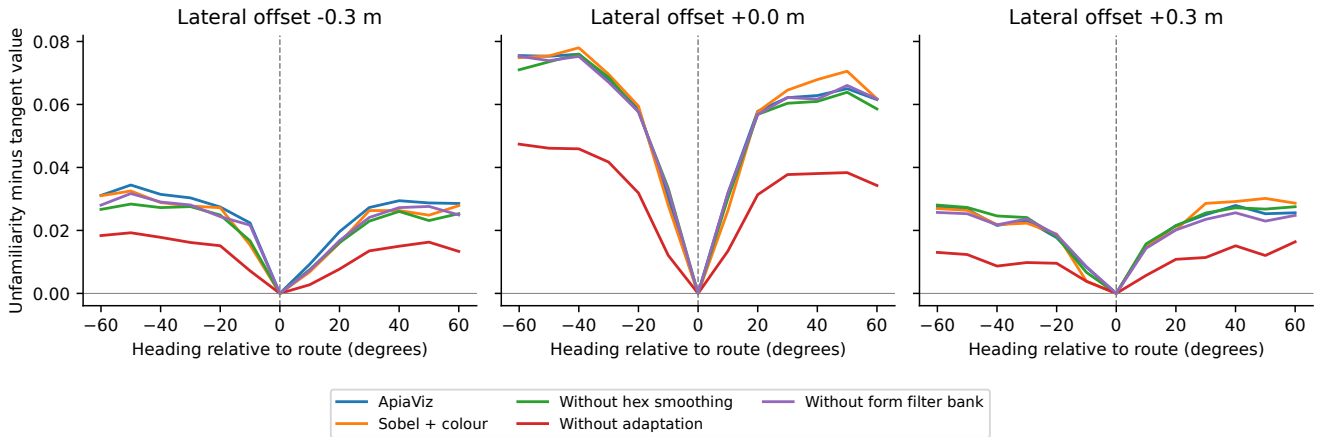


Figure 3: Recognising the route direction is not sufficient for recovery. Lower values are more familiar. Averaged over routes and seeds, familiarity remains greatest near the tangent even 30 cm to either side. These averages can hide individual errors; actual selected turns are summarised below.

At displaced probes, ApiaViz selected inward, parallel and outward turns 35.5%, 39.7% and 24.8% of the time. Sobel gave 35.1%, 40.9% and 24.0%. Learning parallel views across a corridor can therefore preserve the forward direction without supplying a reliable inward correction. This matches a known limitation of visual-compass navigation [4]. It is a plausible recovery mechanism, not proof that the controller alone explains the observed ranking.

5 Implications

The fixed spiking representation can support navigation, and adaptation clearly changes the information available to memory. The broader claim that ApiaViz is a more accurate visual backbone than simple processing is currently unsupported. The next matched experiment should test route-directed learning and restoring steering for every frontend, followed by evaluation in independent worlds with a fixed sample budget. Separately removing the smoothed and ON/OFF channels would better identify which form computations matter.

All 440 trajectories, 61,091 steering decisions and 35,200 probe decisions passed reconstruction checks. The faster renderer matched 560 sampled original images and all four earlier ApiaViz paths and steering scores exactly. These checks do not replace independent-world validation. Other limits are the five seeds, previously selected ApiaViz neural settings, and a full template memory rather than a compact mushroom-body output circuit.

Selected references. Full methods, data, audits and the extended review accompany this note in docs/mechanism-study/.

- [1] Ardin et al. (2016). *PLoS Comput. Biol.* 12:e1004683. doi:10.1371/journal.pcbi.1004683.
- [2] Gattaux et al. (2025). *Nature Communications*. doi:10.1038/s41467-025-62327-3.
- [3] Jesusanmi et al. (2024). *Front. Physiol.* 15:1379977. doi:10.3389/fphys.2024.1379977.
- [4] Amin, Philippides & Graham (2025). *PLoS Comput. Biol.* 21:e1012798. doi:10.1371/journal.pcbi.1012798.
- [5] Owen (2007). The pigeonhole bootstrap. *Ann. Appl. Stat.* 1:386–411. doi:10.1214/07-AOAS122.